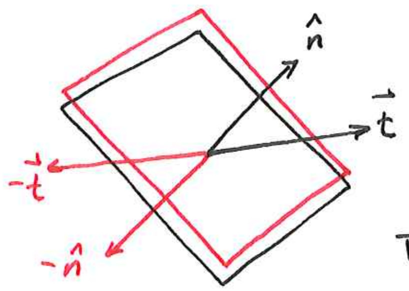
Stress tensor



This plane is arbitrary in space with a normal defined by $\hat{n}$.

The traction (pulls opposite side to interface)

$\vec{t}$ convention:

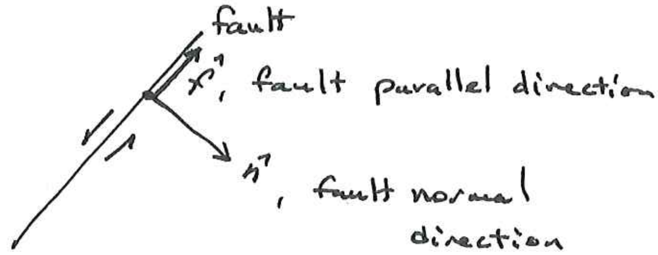
	Seismic	Structural
	extension +	extension −
	compression −	compression +

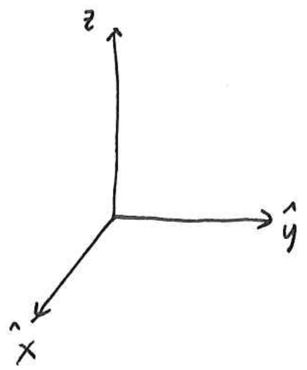
$$\vec{t}(\hat{n}) = (t_x, t_y, t_z)$$

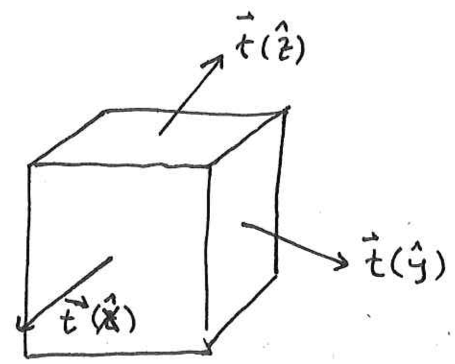
on a fault,

$\vec{t} \cdot \hat{n} = \sigma_n$ normal stress

$\vec{t} \cdot \hat{f} = \sigma_s$ shear stress



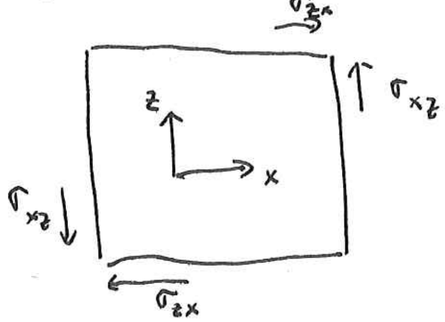




$$\sigma = \begin{bmatrix} t(\hat{x})_x & t(\hat{x})_y & t(\hat{x})_z \\ t(\hat{y})_x & t(\hat{y})_y & t(\hat{y})_z \\ t(\hat{z})_x & t(\hat{z})_y & t(\hat{z})_z \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

stress is symmetric, $\sigma^T = \sigma$

no torque, = no rotation



$\sigma_{xz} = \sigma_{zx}$

$\sigma_{yx} = \sigma_{xy}$

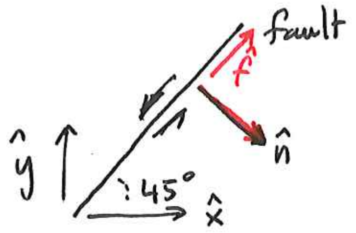
$\sigma_{zy} = \sigma_{yz}$

To find traction in any arbitrary direction

$$\vec{t}(\hat{n}) = \vec{\sigma}\,\hat{n} = \begin{bmatrix} t_x(\hat{n}) \\ t_y(\hat{y}) \\ t_z(z) \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} \hat{n}_x \\ \hat{n}_y \\ \hat{n}_z \end{bmatrix}$$

Suppose we have the 2-D ~~[illegible]~~ stress

$$\vec{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix} = \begin{bmatrix} -40 & -10 \\ -10 & -60 \end{bmatrix} \text{ MPa}$$



$$\hat{n} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} \qquad \hat{f} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

~~normal~~ traction

$$\vec{t}(\hat{n}) = \vec{\sigma}\,\hat{n} = \begin{bmatrix} -40 & -10 \\ -10 & -60 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} -20\sqrt{2} + 5\sqrt{2} \\ -5\sqrt{2} + 30\sqrt{2} \end{bmatrix} = \begin{bmatrix} -15\sqrt{2} \\ 25\sqrt{2} \end{bmatrix} \text{ MPa}$$

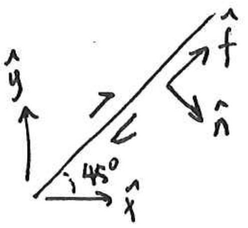
normal traction

$$t_N = \vec{t}\cdot\hat{n} = (-15\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + \left(25\sqrt{2}\right)\frac{\sqrt{2}}{2} = 10 \text{ MPa}$$ + = extension

shear traction

$$t_s = \vec{t}\cdot\hat{f} = (-15\sqrt{2})\left(\frac{\sqrt{2}}{2}\right) + \left(25\sqrt{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = -50 \text{ MPa}$$ -40

$-$ indicates ~~left~~ right lateral



Principle Axes.

For any $\vec{\sigma}$ we can find an $\hat{n}$ that results in no shear stress, i.e.,

$$\vec{t}(\hat{n}) = \vec{\sigma}\hat{n} = \lambda \hat{n}$$

$$(\vec{\sigma} - \lambda \mathbb{I})\hat{n} = 0 \qquad \text{eigen values } \lambda$$

$\uparrow$ identity matrix

$$\det(\vec{\sigma} - \lambda \mathbb{I}) = 0$$

$$\det \begin{vmatrix} \sigma_{xx}-\lambda & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy}-\lambda & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz}-\lambda \end{vmatrix} = (\sigma_{xx}-\lambda)\left((\sigma_{yy}-\lambda)(\sigma_{zz}-\lambda) - \sigma_{yz}^2\right)$$

$$+ \sigma_{xy}\left((\sigma_{yz}\sigma_{xz}) - \sigma_{xy}(\sigma_{zz}-\lambda)\right)$$

$$+ \sigma_{xz}\left(\sigma_{xy}\sigma_{yz} - (\sigma_{yy}-\lambda)(\sigma_{xz})\right)$$

Third order polynomial

$\Rightarrow$ 3 different $\hat{n}$ and 3 λ

$\lambda_1, \lambda_2, \lambda_3 \Rightarrow \sigma_{11}, \sigma_{22}, \sigma_{33}$

$\hat{n}^{(1)}, \hat{n}^{(2)}, \hat{n}^{(3)}$

if $\sigma_{11} = \sigma_{22} = \sigma_{33} \rightarrow -p$ hydrostatic pressure

or as $(\sigma_{11}+\sigma_{22}+\sigma_{33})/3 = p$

often useful to express as deviatoric stress,

$$\vec{\sigma} = \vec{\sigma}_m \mathbb{I} + \vec{\sigma}_D$$

$$= \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix} + \begin{bmatrix} \sigma_{xx}+p & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy}+p & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz}+p \end{bmatrix}$$

Strain tensor

Consider a point $\vec{r}_0$ in space. At some time later, t, the point is at $\vec{r}$. The displacement is

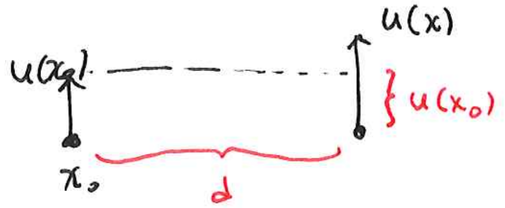
$$\vec{u}(\vec{r}_0, t) = r - r_0$$

$$\vec{u}(t) = \text{particle displacement}$$

$$\frac{\partial \vec{u}(t)}{\partial t} = \text{particle velocity}$$

$$\frac{\partial^2 \vec{u}(t)}{\partial t^2} = \text{particle acceleration}$$

The displacement field at x a distance d from x_0 is



$$\vec{u}(\vec{x}) = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \vec{u}(\vec{x}_0) + \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial z} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix} = \vec{u}(\vec{x}_0) + \mathbb{J} d$$

Jacobian

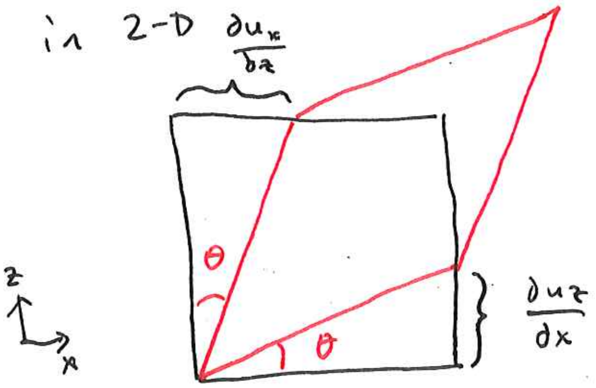
$$\mathbb{J} = \underset{\substack{\text{strain tensor} \\ \text{symmetric}}}{e} + \underset{\substack{\text{rotation} \\ \text{antisymmetric}}}{\Omega}$$

$$\cancel{e = \begin{bmatrix} \frac{1}{2}\left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}\right) \\ \frac{1}{2}\left(\frac{\partial u_z}{\partial x} + \frac{\partial u_x}{\partial z}\right) \end{bmatrix}}$$

$$
e = \begin{bmatrix} \dfrac{\partial u_x}{\partial x} & \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial y} + \dfrac{\partial u_y}{\partial x}\right) & \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial z} + \dfrac{\partial u_z}{\partial x}\right) \\ \dfrac{1}{2}\left(\dfrac{\partial u_y}{\partial x} + \dfrac{\partial u_x}{\partial y}\right) & \dfrac{\partial u_y}{\partial y} & \dfrac{1}{2}\left(\dfrac{\partial u_y}{\partial z} + \dfrac{\partial u_z}{\partial y}\right) \\ \dfrac{1}{2}\left(\dfrac{\partial u_z}{\partial x} + \dfrac{\partial u_x}{\partial z}\right) & \dfrac{1}{2}\left(\dfrac{\partial u_z}{\partial y} + \dfrac{\partial u_y}{\partial z}\right) & \dfrac{\partial u_z}{\partial z} \end{bmatrix}
$$

$$
\Omega = \begin{bmatrix} 0 & \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial y} - \dfrac{\partial u_y}{\partial x}\right) & \dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial z} - \dfrac{\partial u_z}{\partial x}\right) \\ -\dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial y} - \dfrac{\partial u_y}{\partial x}\right) & 0 & \dfrac{1}{2}\left(\dfrac{\partial u_y}{\partial z} - \dfrac{\partial u_z}{\partial y}\right) \\ -\dfrac{1}{2}\left(\dfrac{\partial u_x}{\partial z} - \dfrac{\partial u_z}{\partial x}\right) & -\dfrac{1}{2}\left(\dfrac{\partial u_y}{\partial z} - \dfrac{\partial u_z}{\partial y}\right) & 0 \end{bmatrix}
$$

in 2-D



generally, we assume $\Omega = 0$
no rotation.

The volume change (increase = dilation)

$$
\Lambda = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} = \mathrm{trace}(e) = \nabla \cdot \vec{u}
$$

we can find principal strain directions similar to stress

$$
\vec{u} = \vec{e}\hat{n} = \lambda \hat{n}
$$

eigen values, e_1, e_2, e_3

notation each strain component, e_{ij}

$$e_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$$

if $i = x$ $j = y$.

$$e_{xy} = \frac{1}{2}\left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y}\right)$$

Generalized stress - strain relationship.

$$\sigma_{ij} = \underbrace{c_{ijkl}}_{\text{elastic tensor}} e_{kl} \equiv \sum_{k=1}^{3}\sum_{l=1}^{3} c_{ijkl} e_{kl}$$

$1 = x$ $2 = y$ $3 = z$

elastic tensor, has 81 components

If isotropic solid and invariant to rotation, reduces to 2 independent parameters

$$c_{ijkl} = \lambda \delta_{ij}\delta_{kl} + \mu(\delta_{il}\delta_{kl} + \delta_{ik}\delta_{jl})$$

Lamé constants

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$c_{1111} = \lambda + 2\mu$

$c_{1112} = 0$

$c_{1122} = \lambda$

$c_{1212} = \mu$

$$\sigma_{ij} = [\lambda \delta_{ij}\delta_{kl} + \mu(\delta_{ij}\delta_{jk} + \delta_{ik}\delta_{jl})] e_{kl}$$

It can be shown that

$$\sigma_{ij} = \lambda \delta_{ij} e_{kk} + 2\mu e_{ij} \qquad \text{where } e_{kk} = tr(\vec{e})$$

$$\sigma = \begin{bmatrix} \lambda tr(e) + 2\mu e_{11} & 2\mu e_{12} & 2\mu e_{13} \\ 2\mu e_{21} & \lambda tr(e) + 2\mu e_{22} & 2\mu e_{23} \\ 2\mu e_{31} & 2\mu e_{32} & \lambda tr(e) + 2\mu e_{33} \end{bmatrix}$$

$$E = \frac{(3\lambda + 2\mu)\mu}{\lambda + \mu} \qquad K = \lambda + \frac{2}{3}\mu \qquad \text{Poisson's ratio } \nu = \frac{\lambda}{2(\lambda + \mu)}$$

General wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

general solutions $\quad u(x,t) = f(x \pm ct)$

$\uparrow$ general waveform

Homogeneous equation of motion

$$\rho \frac{\partial^2 u}{\partial t^2} = \partial_j \tau_{ij}$$

$$\tau_{ij} = \lambda \delta_{ij} e_{kk} + 2\mu e_{ij}$$

$$e_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$$

$$\tau_{ij} = \lambda \delta_{ij} \partial_k u_k + \mu(\partial_i u_j + \partial_j u_i)$$

eventually yields the generalized seismic wave equation.

$$\rho \frac{\partial^2 \vec{u}}{\partial t^2} = \nabla\lambda(\nabla\cdot\hat{u}) + \nabla\mu\cdot\left[\nabla u + (\nabla u)^T\right] + (\lambda+\mu)\nabla\nabla\cdot\hat{u} + \mu\nabla^2\vec{u}$$

$$\vec{t}(\hat{n}) = \vec{\sigma}\hat{n} = \begin{bmatrix} t_x(\hat{n}) \\ t_y(\hat{n}) \\ t_z(\hat{n}) \end{bmatrix} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} \hat{n}_x \\ \hat{n}_y \\ \hat{n}_z \end{bmatrix}$$

$$\rho \frac{\partial^2 u}{\partial t^2} = \underbrace{\nabla\lambda(\nabla\cdot\vec{u}) + \nabla\mu\cdot[\nabla\vec{u} + (\nabla\vec{u})^T]}_{\nabla\lambda \text{ and } \nabla\mu \neq 0 \text{ when Earth is inhomogeneous}} + \underbrace{(\lambda+2\mu)\nabla\nabla\cdot\vec{u} - \mu\nabla\times\nabla\times\vec{u}}_{P \,\&\, S \text{ waves}}$$

To solve, assume $\nabla\lambda$ and $\nabla\mu = 0$ in thin layers, but change between layers.

$$\rho \frac{\partial^2 u}{\partial t^2} = (\lambda+2\mu)\nabla\nabla\cdot\vec{u} - \mu\nabla\times\nabla\times\vec{u}$$

Break into two parts

$$\nabla\cdot(\nabla\times\Psi) = 0$$

$$\nabla\cdot\left(\rho \frac{\partial^2 u}{\partial t^2} = (\lambda+2\mu)\nabla\nabla\cdot\vec{u} - \mu\nabla\times\nabla\times\vec{u}\right)$$

$$\rho \frac{\partial^2(\nabla\cdot u)}{\partial t^2} = (\lambda+2\mu)\nabla^2(\nabla\cdot\vec{u}) - \mu\left(\cancelto{0}{\nabla\cdot\nabla\times\nabla\times\vec{u}}\right)$$

$$\frac{\rho}{\lambda+2\mu}\frac{\partial^2(\nabla\cdot u)}{\partial t^2} = \nabla^2(\nabla\cdot\vec{u}) \qquad \alpha = V_p = \left(\frac{\lambda+2\mu}{\rho}\right)^{1/2}$$

$$\frac{\partial^2(\nabla\cdot u)}{\partial t^2} = \alpha^2\nabla^2(\nabla\cdot\vec{u})$$

Instead, taking $\nabla\times(\nabla\phi)$

$$\nabla\times\left(\rho\frac{\partial^2\vec{u}}{\partial t^2} = (\lambda+2\mu)\nabla\nabla\cdot\vec{u} - \mu\nabla\times\nabla\times\vec{u}\right)$$

$$\rho\frac{\partial^2(\nabla\times\vec{u})}{\partial t^2} = (\lambda+2\mu)\cancelto{0}{\nabla\times\nabla\nabla\cdot\vec{u}} - \mu\nabla\times\nabla\times(\nabla\times\vec{u})$$

$$\nabla\times\nabla\times\vec{u} = \nabla\nabla\cdot\vec{u} - \nabla^2\vec{u}$$

$$\rho\frac{\partial^2(\nabla\times\vec{u})}{\partial t^2} = \cancel{(\lambda+2\mu)} - \mu\nabla\times(\nabla\nabla\cdot\vec{u} - \nabla^2\vec{u})$$

$$= -\mu\,\cancelto{0}{\nabla\times\nabla\nabla}\cdot\vec{u} + \mu\,\cancel{\nabla\times}\nabla^2(\nabla\times\vec{u})$$

$$\frac{\partial^2(\nabla\times\vec{u})}{\partial t^2} = \frac{\mu}{\rho}\nabla^2(\nabla\times\vec{u}) \qquad \beta = V_\rho = \left(\frac{\mu}{\rho}\right)^{1/2}$$

$$\frac{\partial^2(\nabla\times\vec{u})}{\partial t^2} = \beta\nabla^2$$

$$\frac{\partial\vec{u}}{\partial t^2} = \alpha^2\nabla\nabla\cdot\vec{u} - \beta^2\nabla\times\nabla\times\vec{u}$$

$$\vec{u} = \nabla\phi + \nabla\times\psi \qquad \nabla\cdot\psi = 0$$

$$\nabla\cdot\vec{u} = \nabla^2\phi$$

$$\nabla\times\vec{u} = \nabla\times\nabla\times\psi$$